HW 4.3: Reproducible Data Analysis

Stat 184

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0.1 Busiest Airport Analysis

We looked up and found some of the six busiest airports around the world from 2020-2025. We then compare the number of passengers yearly to each other. See this information presented in the table and plots below.

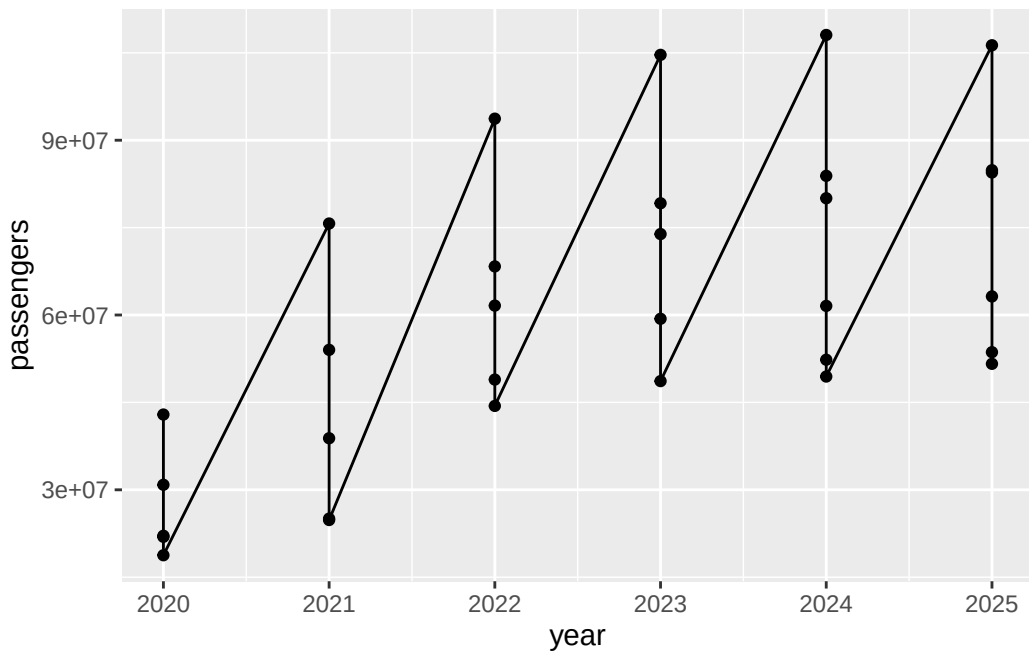
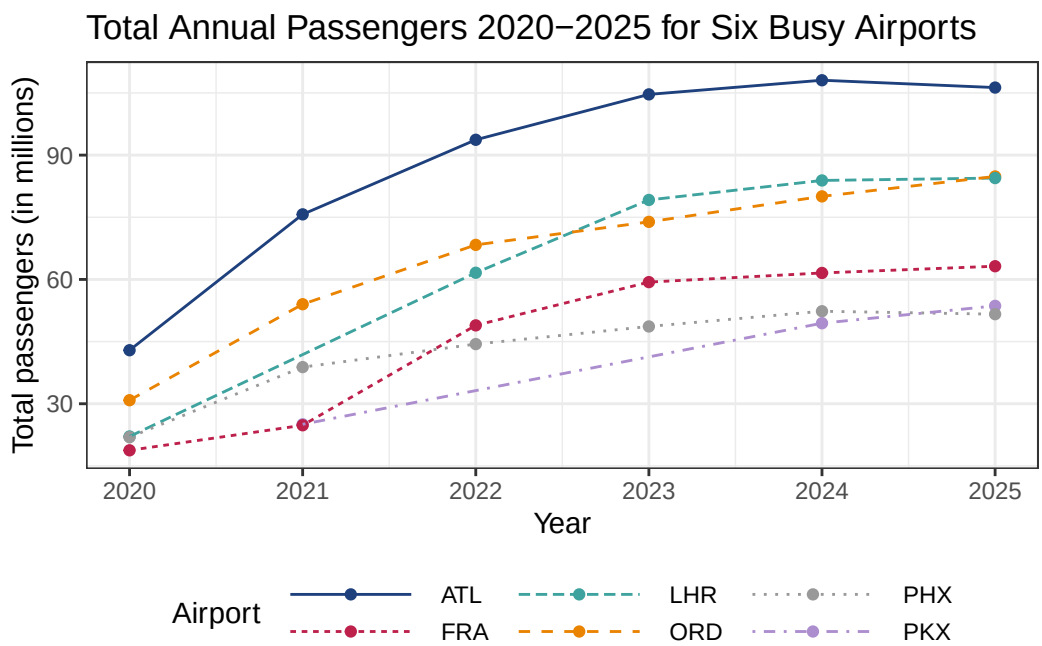


Table 1: Total Annual Passengers 2020-2025 for Six Busy Airports

Airport	IATA	2020	2021	2022	2023	2024	2025
Hartsfield–Jackson Atlanta International Airport	ATL	42,918,685	75,704,760	93,699,630	104,653,451	108,067,766	106,302,208
O'Hare International Airport	ORD	30,860,251	54,020,399	68,340,619	73,894,226	80,043,050	84,851,825
Heathrow Airport	LHR	22,111,469		61,614,508	79,183,364	83,884,572	84,463,000
Phoenix Sky Harbor International Airport	PHX	21,928,708	38,846,713	44,397,854	48,654,432	52,325,266	51,620,420
Frankfurt Airport	FRA	18,768,601	24,812,849	48,918,482	59,355,389	61,564,957	63,189,666
Beijing Daxing International Airport	PKX		25,051,012			49,441,029	53,618,949

Note: Airports that were not in the top 50 have missing passenger data; highlighted rows are the three required airports.



The table shows the annual passenger counts for Hartsfield-Jackson Atlanta, O'Hare, Heathrow, Phoenix Sky Harbor, Frankfurt, and Beijing Daxing International Airport. We can see that the busiest airport in terms of passengers is Hartsfield-Jackson Atlanta. We can also find from the table that there were some years for Beijing and Heathrow where they were not within the top 50 busiest airports.

The plots do a great job of showing the gradual increase in total annual passengers. Due to the pandemic, airports are shutting down and slowly opening back up over the years.

0.2 Monte Carlo Numerical Integration

Numerical integration refers to the process of evaluating a definite integral and getting a number. In most calculus classes, students learn deterministic algorithms such as Riemann Sums or the Trapezoidal Rule. However, we can also use *stochastic* algorthims.

1. Identify the function you want to integrate; we'll use

$$f(x) = -(x-4)^2 + 4$$

2. Identify the bounds of your definite integral; we'll use

$$2 \leq x \leq 6$$

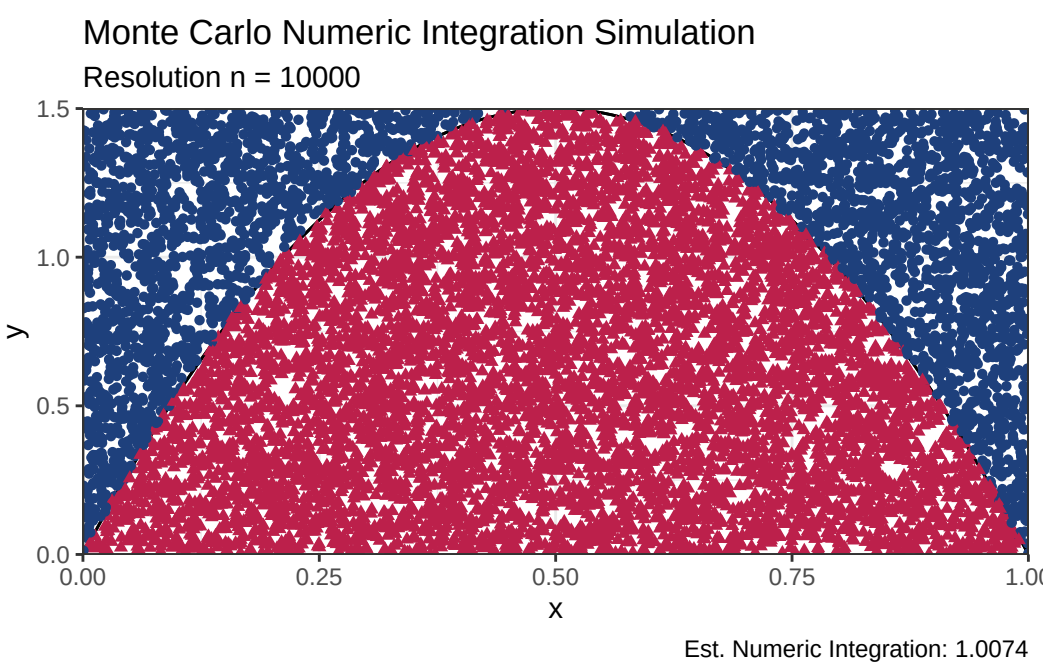
3. Plot your function over the bounds of integration to determine the bounds on the output y This will let us create a rectangular window to focus on. We'll use

$$0 \leq y \leq 4$$

4. Find the area of your rectangular window (we will need this later):

$$(6-2)(4-0) = 16$$

5. Set up the Monte Carlo Simulation:
   1. Randomly sample n x-coordinates using a Uniform distribution (e.g., using `runif` in R) from the interval
   $$2 \leq x \leq 6$$
   2. Randomly sample n y-coordinates using a Uniform distribution from the interval
   $$0 \leq y \leq 4$$
   3. Join the two samples together into one data frame
6. Identify whether each (x, y) ordered pair is above or below the function; that is, check whether
   $$y > f(x)$$
   or
   $$y \leq f(x)$$
7. Calculate the proportion of points that are on or below the graph of the function, say p.
8. Multiply the proportion p by the area of the rectangular window to get an estimate for the integral;
   $$16 \cdot p$$



This plot shows a Monte Carlo simulation used to estimate the value of a definite integral for the Beta(2,2) density over the interval
$$0 \leq x \leq 10$$
. A total of 10,000 points were randomly generated within the rectangular region defined by
$$0 \leq x \leq 1$$

and

$$0 \leq y \leq 1.5$$

, which fully contains the curve. Each point is classified based on whether it lies above or on/below the function, with points under the curve forming the shaded region that represents the area of interest. The proportion of points that fall on or below the curve is used to approximate the integral by multiplying this proportion by the area of the rectangle. The estimate shown, approximately 1.0074, is very close to the true value of 1, demonstrating that the Monte Carlo method provides an accurate approximation when a large number of random samples is used.

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

0.3 Coding/AI Apendix

Used chatgpt for monte carlo long text: This plot shows a Monte Carlo simulation used to estimate the value of a definite integral for the Beta(2,2) density over the interval 0 x 10 x 10 x 1. A total of 10,000 points were randomly generated within the rectangular region defined by 0 x 10 x 10 x 1 and 0 y 1.50 y 1.50 y 1.5, which fully contains the curve. Each point is classified based on whether it lies above or on/below the function, with points under the curve forming the shaded region that represents the area of interest. The proportion of points that fall on or below the curve is used to approximate the integral by multiplying this proportion by the area of the rectangle. The estimate shown, approximately 1.0074, is very close to the true value of 1, demonstrating that the Monte Carlo method provides an accurate approximation when a large number of random samples is used.

Airport Code:

1 Step 1: Load Packages —-

library(rvest) library(tidyverse) # loads {dplyr}, {tidyr}, {ggplot2} library(knitr) library(kableExtra)

2 Step 2: Get the Data —-

2.1 Step 2a: Scrape the data; 2020-2025

airportWikiData <- read_html(x = "https://en.wikipedia.org/wiki/List_of_busiest_airports_by_passenger_traffic") %>% html_elements(css ="table") %>% html_table()

2.2 Step 2b: Extract data and add year columns

airports_2025 <- airportWikiData[[2]] %>% mutate(year = 2025, .before = Rank)

airports_2024 <- airportWikiData[[3]] %>% mutate(year = 2024, .before = Rank)

airports_2023 <- airportWikiData[[4]] %>% mutate(year = 2023, .before = Rank)

airports_2022 <- airportWikiData[[5]] %>% mutate(year = 2022, .before = Rank)

airports_2021 <- airportWikiData[[6]] %>% mutate(year = 2021, .before = Rank)

airports_2020 <- airportWikiData[[7]] %>% mutate(year = 2020, .before = Rank)

2.3 Step 2c: Combine data frames & 2e: filter to target airports

2.3.1 Split Airport codes to separate columns

2.3.2 Fix column names and convert passengers, rank change, & percent change to numbers

targetAirports <- c("ATL", "FRA", "PKX", "PHX", "ORD", "LHR")

airportData6 <- bind_rows(airports_2020, airports_2021, airports_2022, airports_2023, airports_2024, airports_2025) %>% rename(# Fix column names passengers = Totalpassengers, percentChange = `%change`) %>% separate_wider_delim(# Split airport codes apart cols = `Code(IATA/ICAO)`, delim = "/", names = c("IATA", "ICAO"), too_few = "align_start", too_many = "merge") %>% filter(IATA %in% targetAirports) %>% mutate(# Convert numbers to numbers across(.cols = c(passengers, percentChange), .fns = ~parse_number(.x)))

2.4 OPTIONAL: Clean up excess data tables

remove(list = c("airports_2020", "airports_2021", "airports_2022", "airports_2023", "airports_2024", "airports_2025", "airportWikiData"))

3 Busiest Passenger Airports —-

3.1 Create two data visualizations (one table and one plot) for six

3.2 of the busiest passenger airports in the world, across the

3.3 years 2020-2025.

4 NOTE: This example picks up from the previous example.

5 Step 3: Make a Table —-

5.1 Step 3a: Subset data to airport, year, and passenger total

5.2 Step 3b: Pivot so case is an airport instead of airport-year

airportData6_wide <- airportData6 %>% dplyr::select(year, Airport, IATA, passengers) %>% pivot_wider(names_from = year, values_from = passengers)

5.3 Step 3c: Improve/polish cycle

5.3.1 Add commas, title, set NA to blanks, center-align numbers

5.3.2 Highlight 3 required airports

options(knitr.kable.NA = " ") airportData6_wide %>% kable(caption ="Total Annual Passengers 2020-2025 for Six Busy Airports", align = "lcccccc", format = "latex", format.args = list(big.mark = ",")) %>% kable_classic(full_width = FALSE) %>% kable_styling(latex_options = "scale_down") %>% row_spec(row = c(1, 5, 6), background = "#FFFFC5") %>% footnote(general = "Airports that were not in the top 50 have missing passenger data; highlighted rows are the three required airports.", footnote_as_chunk = TRUE)